The 20-Round Barrier for Table-Based Preimage Attacks on SHA-256

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Abstract

This is a companion note to [1], which gives a practical preimage attack on the 19-round SHA-256 compression function built on a precomputed table inverting $u \mapsto \sigma_0(u) - u$. Here we ask why the same construction does not reach 20 rounds, and characterise the barrier quantitatively rather than merely reporting it.

The R=19 attack is shown not to schedule its constraints acyclically: it breaks a cycle by freezing W_9 provisionally, iterating, and paying one self-consistency check. Promoting a_{11} to an unknown absorbed by the fourth constraint gives R=20 the same one-word surplus, and the true R=20 solution is verified to be a fixed point of all four absorber maps—so the architecture transplants in form. It does not transplant in efficacy: measured with a single harness, the R=19 two-map iteration converges at 5.34×10^{-5} per a_0 (1,790 of 3.36×10^7) while the R=20 four-map iteration yields 0 of 2.00×10^9 ($\leq 1.51 \times 10^{-9}$ at 95%), below the 2^{-29} break-even. The cause is dimensional: the iterated state grows from 32 to 64 bits.

We then give the structural reason. Each free state word a_k enters exactly two recovered message words linearly, W_k with coefficient +1 and W_{k+8} with coefficient −1, so constraint C_j absorbs a_{j+1} through the one global $\sigma_0(u)-u$ table for *every* j, C3→ a_4 included. At R=19 dependence among the absorbed words is full-avalanche only in the forward direction and every feedback edge passes through Maj/Ch alone (at most 1.9 bits of a constraint change per flipped bit); at R=20, in the frame that absorbs a_4, exactly one feedback edge is heavy, a_4 →C0 through W_9 (12.2 bits), and in the frame that absorbs a_{11} three are; either way the heavy dependency graph acquires its first cycle. The R=19 iteration works because it cuts its cycle at a mild edge; a heavy cycle has no mild cut, and any 32-bit cut of it behaves, as far as we can measure, like a random map. The obstruction is $\Sigma_0(a_4)$ subtracted directly inside W_9, which no context choice removes; it is the reach of the round function, not of the frame: the window is all-mild exactly when $R \leq 19$, and R=21 has three heavy feedback edges. We give the corrected inventory of which absorptions admit a global table, show that with the tables we know the assignment is forced up to a choice between two frames, both of which close the heavy cycle, and record the resulting cost: 2^{64} per context-equivalent, i.e. 2^{32} times the R=19 per-context cost, for this attack family. We record a second global table (the a_0 absorber, $\Gamma(a_0) = a_0 + g(a_0)$, coverage 63.214% against $1 - e^{-1} = 63.212\%$), and prove that $\mathrm{rank}(\sigma_0 \oplus I) = 31$ gives a unique mask $\lambda = \texttt{0xa8a42fe4}$ with $\lambda \cdot (\sigma_0(W) \oplus W) = 0$ for every W (verified exhaustively over all 2^{32} inputs), which modular subtraction degrades to a bias of only $2^{-15.8}$.

Because these are properties of the parameterisations we tried rather than proofs of impossibility, we also build a reduced-width scale model in which the whole inner space is enumerable, and report exhaustive measurements there. **None of this is an impossibility result**, and we are explicit throughout about which claims are algebraic and which are statistical models that remain untested at full width. Full SHA-256 (64 rounds) is not threatened by any of these results.

Keywords: SHA-256, preimage attack, reduced-round, round-20 barrier, absorber tables, reduced-width scale model.

1 Scope and Relation to the $R = 19$ Paper

We assume the attack of [1] throughout: a backward chain from the target fixes $a_{12}, \dots, a_{19}$ (for R=20), the context words $a_4, \dots, a_{10}$ are freely chosen (one of the two admissible frames, §3.2), and the remaining unknowns are resolved against the schedule-consistency constraints C0–C3 at $W_{16}, \dots, W_{19}$ using a 16 GB global table inverting $u \mapsto \sigma_0(u) - u$.

Notation. We follow [1]: $T_1^{(r)}$, $T_2^{(r)}$ and e_r are the round quantities of round r; $g(a_0) = W_1 - a_1$ is the part of W_1 that depends only on a_0, as defined there; $D(a_3)$ is the a_3-dependent part of W_{10} and F_{12}, F_{23} the (a_0, a_1)- and (a_0, a_1, a_2)-dependent parts of W_2 and W_3, as in its Proposition 3; and $c_0 = W_0 - a_0$ is the round-0 constant (its C_0). Constants that depend only on the context and the target are written CONST with a subscript. R_r for $r = 16, \dots, 19$ denotes the residual of constraint C_{r-16}: the message word W_r recovered from the state minus its schedule value, zero exactly when the constraint holds. $H(a_{11})$ denotes the right-hand side of C3 as a function of a_{11} alone and $h_1(a_{11})$ the a_{11}-dependent part of the C1 target. A *frame* is a choice of which state words are context and which are absorbed (§3.2).

Scope. "Barrier" here is specific to this attack family. Meet-in-the-middle and biclique preimage attacks reach 43 and 45 steps at costs near 2^{256} [2, 3], and Zaikin [4] has inverted the 19-round compression function for a fixed digest by parameterized Cube-and-Conquer; nothing here bears on those results.

Nothing here modifies the R=19 result. The material is separated because it is exploratory: it consists largely of negative results, and its value is in narrowing where a R=20 attack could come from, not in delivering one.

2 The $R = 20$ Constraint System

The backward chain at $R = 20$ recovers $a_{12}, \dots, a_{19}$ but not a_{11}. The schedule grows to four constraints W_{16}–W_{19}, with a_{11} as an additional unknown.

Reference-conditioned result. When the context including a_{11} is taken from a known preimage of the target, the W_{17} identity gives a_2 as an explicit function of a_3 (Lemma analogous to [1, Prop. 3]), and an h-table $h[W_{12}(a_{11})] \to a_{11}$ enables an $O(2^{32})$ a_3 sweep. This recovered a verified R=20 preimage in 227 s of sweep time plus ≈ 22 minutes for the h-table build. However, the context was derived from a known preimage: this is a second preimage computed with knowledge of the first, not a preimage attack in the standard sense.

Why the R=20 preimage attack is not established. The h-table inversion requires $W_{12}^{\mathrm{req}} = W_{19}^{\mathrm{bc}} - \sigma_1(W_{17}^{\mathrm{bc}}) - \sigma_0(W_4) - W_3$, where W_{17}^{bc} carries a_{11} through e_{13} (which contains $\mathrm{Maj}(a_{12}, a_{11}, a_{10})$). Thus a_{11} is needed to compute the quantity used to look up a_{11}: a circular dependency. No algebraic link that breaks this circularity has been found. Exhaustive search over all binary linear combinations of the four residual equations R_{16}–R_{19} found no combination that isolates any of the three unknowns (a_2, a_3, a_{11}).

Quantitative complexity gap. An equivalent view using the σ_0-chain directly confirms the 2^{32} difficulty increase. In the classical accounting, refined in §3.2, treat $a_4, \dots, a_{11}$ as freely chosen context (the backward chain fixes $a_{12}, \dots, a_{19}$). The chain C0–C2 then proceeds identically to the $R = 19$ case, but the schedule now includes W_{19}, introducing a fourth constraint (C3):

$$\sigma_0\big(W_4^{\mathrm{round}}\big) = W_{19}^{\mathrm{bc}} - \sigma_1(W_{17}^{\mathrm{bc}}) - W_{12}^{\mathrm{base}} - W_3,$$

where W_4^{round} is fully determined by $a_0, \dots, a_3$ once C0–C2 converge and W_{12}^{base} is determined by context. Because σ_0 is a bijection, W_4^{round} must equal the unique $\sigma_0^{-1}(C3_{\mathrm{tgt}})$. C3 therefore contributes an additional 32-bit equality. Under the present construction we model it as approximately uniform and independent of C0–C2, so that it passes with probability $\approx 2^{-32}$; we stress that this is a statistical model which has not been established directly at full width.

Exhaustive experiments in a reduced-width scale model (§2.1) bear on the model but do not confirm it at $w = 32$. Measuring $P(\mathrm{C3} \mid \mathrm{C0}, \mathrm{C1}, \mathrm{C2})$ against the independent prediction 2^{-w} gives a ratio of 1.075 ± 0.018 at $w = 4$ and 1.005 ± 0.035 at $w = 5$: consistent with exact independence at the larger width, with a small excess at the smaller one that may be a narrow-width artefact. Any dependence is thus small, but two widths do not establish a trend, and the behaviour at $w = 32$ remains untested.

Schedule identity. For messages satisfying the σ_0-chain $W_k = \sigma_0(W_{k+1})$, $k = 0, \ldots, 3$, the substitution $\sigma_0(W_{r-15}) = W_{r-16}$ reduces the schedule recurrence for $r \in \{16, \ldots, 19\}$ to

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + 2\, W_{r-16}.$$

From this identity and $W_3 = \sigma_0(W_4)$ one computes directly $C3_{\mathrm{tgt}} = W_{19} - \sigma_1(W_{17}) - W_{12} - W_3 = \sigma_0(W_4) = W_3$, confirming that the C3 check reduces to $W_4^{\mathrm{round}} = W_4^{\mathrm{chain}}$ (verified numerically). The same identity provides a concise derivation of the Nine-Step Lemma: at $r = 16$, $\Delta W_{16} = \Delta W_9 + \Delta W_0 = -\delta + \delta = 0$, and the doubled term $2W_{r-16}$ ensures that *any* perturbation $\Delta W_1 = \varepsilon$ would introduce $\sigma_0(\varepsilon)$ into ΔW_{16} (breaking cancellation), which is why the pair $(0, 9)$ is the unique minimal solution.

Orbit structure of $\boldsymbol{\sigma_0}$. Viewed as a $\mathbb{F}_2$-linear bijection on $(\mathbb{Z}/2\mathbb{Z})^{32}$, the map σ_0 has order $49{,}146 = 2 \cdot 3 \cdot 8191$ where $8191 = 2^{13} - 1$ is a Mersenne prime. Its only short orbits are the two fixed points $\{0,$ `0x27f42515`$\}$ and the unique 2-cycle $\{$`0x0b3b7ef9` $\leftrightarrow$ `0x2ccf5bec`$\}$; in particular, no period-3 or period-4 elements exist. Consequently any σ_0-chain of length four $(W_0, \ldots, W_4)$ lies in an orbit of length 49,146 unless W_4 is one of the four exceptional values above, so no "chain-wrap" shortcut of the form $W_0 = W_4$ arises generically.

Under the variable ordering and single-representative lookup construction used here, C3 remains an additional 32-bit consistency condition applied after C0–C2 have been solved. We emphasise what is and is not established. The algebra shows C3 imposes a further 32-bit equality; it does not show that every algebraic arrangement must pay 2^{32} for it. No absorption of C3 into the construction, and no lower-width decomposition of it, is currently known—but the negative results below are properties of the parameterisations we tried, not proofs of impossibility. Indeed one apparent obstruction at R=20, the $a_1 \leftrightarrow a_2$ cycle, dissolved entirely under a change of variable (§2.2), which is direct evidence that such obstructions can be artefacts of representation. Treating C3 as an independent filter gives the cost in one convention, which we use throughout. An R=19 attempt costs 2^{32} table lookups and succeeds with probability $\mu_{R=19}$, measured at 1.25×10^{-2} and projected at 6×10^{-3}–1.3×10^{-2} [1, §5.6]. An R=20 attempt in the classic harness guesses one further word and costs 2^{64}, so we say the barrier is 2^{64} *per context-equivalent*. The expected number of R=19-sized contexts is then $\mu_{R=19}^{-1} \times 2^{32} \approx 2^{39.4}$ at the projected rate, and the expected total work $\mu_{R=19}^{-1} \times 2^{64} \approx 2^{71.4}$. The factor of 2^{32} between the two round counts is what matters and is independent of $\mu_{R=19}$; the absolute figures inherit its uncertainty.

GPU diagnostic. A 165-context GPU run of the $R = 20$ σ_0-chain on an H100 (batch $= 2^{26}$, $\mathrm{K}_{\mathrm{seeds}} = 4$; script `r20_oracle_free.py`, 19 June 2026, a separate run from the 165-context R=19 run reported in [1]) recorded 4,166 W_9-lo-pass events and zero hi-pass events. Neither this log nor that of the CPU sweep below is included in the artefact package. Under the null hypothesis of uniform residuals, the expected hi count is $\mu = 4166/65536 \approx 0.064$; the probability of zero hi events is $e^{-\mu} \approx 0.94$, so the absence of hi events is unsurprising. With an expectation below 0.1, however, observing zero carries almost no information: the result is *consistent with* a uniform 2^{-32} filter but does not test it. We attribute the difficulty increase over $R = 19$ to the additional C3 equality rather than to any structural bias in W_9; the attribution of the full 2^{32} factor to C3 rests on the uniformity model stated above, which this run is too small to test. A supplementary

CPU sweep (10 contexts, batch = 2^{24}, two workers; script `r20_corr_parallel.py`, 20 June 2026) recorded 254 W_9-lo-pass events and zero C3-pass events. With zero events the one-sided 95% Poisson bound is 3.0, so $P(\text{C3} \mid \text{lo}) \leq 3/254$ and the enhancement over a uniform 2^{-32} filter is at most $3 \cdot 2^{32}/254 = 2^{25.6}$. This too is consistent with a uniform filter and far too small a sample to test independence or estimate its probability. The structural claim—that C3 imposes an additional 32-bit equality—rests on the algebra, not on these counts; the probabilistic claim should be read as untested.

2.1 A Reduced-Width Scale Model

Claims about C3's independence, and about whether the chained solver is lossy, are untestable at full width: the events are too rare to sample and the solution set cannot be enumerated. We therefore build a scale model of the round function and message schedule at word width w, where the inner space $(a_0, a_1, a_2, a_3, a_{11})$ is small enough to enumerate exhaustively. Rotation and shift amounts are scaled from SHA-256's by $w/32$ and rounded; K and the IV are truncated. The construction preserves the shape of the mixing and is not proposed as a standard.

Two fidelity checks are passed. At $w = 32$ the scaled constants reduce to SHA-256's exactly, and the model reproduces SHA-256's state evolution word for word. The solution count per context matches the degrees-of-freedom prediction: five unknown words less four constraints leaves one surplus word, hence 2^w solutions, against 19.2 measured at $w = 4$ ($2^4 = 16$) and 32.0 at $w = 5$ ($2^5 = 32$).

The independence measurement is reported in §2. The model also shows that the chained fixed-point architecture is intrinsically lossy rather than merely slow. A solver using one representative per table entry recovers 1.3%, 0% and 0% of the true solutions at $w = 4, 5, 6$, while one branching over all roots recovers 23.4%, 6.8% and 0.7%. Raising the beam from 64 to 1024 and the iteration count from 6 to 20 changed coverage not at all, so the loss is structural rather than a truncation effect. Tables should therefore be relations, not functions. This monotone collapse, resting on 36, 13 and 2 recovered solutions, is the model's robust finding.

We deliberately do *not* extrapolate these rates to $w = 32$. The per-a_0 success rate falls by 4.15× from $w = 4$ to $w = 5$ and by 8.67× from $w = 5$ to $w = 6$, so the point estimates suggest an accelerating rather than log-linear decay; but the $w = 6$ figure rests on two events, whose exact Poisson interval puts the second ratio anywhere in $[2.40\times, 71.6\times]$, a range that contains the first. The data therefore constrain neither the functional form nor the rate, while a 5% change in the fitted rate, together with the accompanying shift of the $w = 5$ base point, moves a 27-bit extrapolation by a factor of 4.5 (3.8 from the rate alone). The scale model's value lies in its exhaustive in-range comparisons, not in extrapolated rates.

2.2 Absorber Reparameterisation and the $a_1 \leftrightarrow a_2$ Coupling

The four residual equations can each be written in the $\sigma_0(u) - u$ form used by the R=19 table. Substituting the round-function expression $W_{10} = \text{CONST}_{10} - D(a_3) - a_2$ into the W_{17} identity and then eliminating a_2 via $a_2 = W_2 - F_{12}(a_1)$ yields:

Lemma 1 (Reparameterisation of C1)**.** *Write $A(a_1) := a_1 + g(a_0) + F_{12}(a_1)$, a function of a_1 alone once a_0 is fixed. Then*

$$\sigma_0(W_2) - W_2 \;=\; h_1(a_{11}) - \text{CONST}_{10} + D(a_3) - A(a_1) \;=:\; u,$$

so the target u decomposes additively into independent contributions from a_{11}, a_3 and a_1.

Verified exactly on 2000/2000 random triples (a_1, a_3, a_{11}). Taking u as the free variable inverts the map: W_2 is one table lookup, $a_1 = A^{-1}(\kappa - u)$ with $\kappa := h_1(a_{11}) - \text{CONST}_{10} + D(a_3)$ a second, and $a_2 = W_2 - F_{12}(a_1)$. Hence for fixed (a_3, a_{11}) each u determines *both* a_1 and a_2 in

$O(1)$. The apparent circular dependency between C0 (which needs a_2) and C1 (which needs a_1) is therefore an artefact of the parameterisation, not a genuine fixed-point problem; this explains why naive iteration of that pair never converges. (A behaves like a random map: 3 collisions in 2×10^5 samples against 4.7 expected for a random map and 0 for a bijection, so A^{-1} is a relation with about 63% coverage and requires a bucketed table.)

Two further reductions place the remaining constraints in table form. In C2 the unknown a_3 cancels from its own target, since $e_7 = a_3 + T_1^{(7)}$ and $W_3 = a_3 + F_{23}$:

$$\sigma_0(W_3) - W_3 = \big(W_{18} - \sigma_1(W_{16}) - W_{11}\big) - W_2 - F_{23} + T_1^{(7)}.$$

In C3, the quantities e_{16}, e_{17}, e_{18} do not depend on a_{11} (they are fixed by the backward chain), so $\sigma_1(e_{18})$ and $\mathrm{Ch}(e_{18}, e_{17}, e_{16})$ are per-context constants and

$$W_{19} \;=\; \mathrm{CONST}_{19} - \big(a_{11} + T_1^{(15)}\big)$$

is *linear* in a_{11}; only W_{17} and W_{12} carry nonlinear a_{11}-dependence into the C3 key.

2.3 A Context Family that Removes a_2 from C0

The variable a_2 reaches the C0 target by exactly two routes: the term $\mathrm{Ch}(e_8, e_7, e_6)$ with $e_6 = a_2 + T_1^{(6)}$, and $\mathrm{Maj}(a_4, a_3, a_2)$ inside $T_1^{(5)}$. Both are maskable by context choice. Setting $e_8 =$ `0xFFFFFFFF` (reachable by choosing a_8, since $e_8 = a_4 + a_8 - \Sigma_0(a_7) - \mathrm{Maj}(a_7, a_6, a_5)$) collapses $\mathrm{Ch}(e_8, e_7, e_6)$ to e_7; setting $a_4 = a_3$ collapses $\mathrm{Maj}(a_4, a_3, a_2)$ to a_3. The second condition ties a context word to an absorbed unknown, so the context constants would have to be rebuilt for every candidate a_3; it is recorded as a structural observation, not as an attack step. Counting distinct C0 targets over several hundred random a_2 per trial: with a random context, or with either condition alone, essentially every a_2 gives a distinct target; with both conditions the count is exactly one in each of four trials.

Neither condition suffices alone; together they eliminate a_2 from C0 entirely, turning it into a two-input absorber $(a_3, a_{11}) \mapsto a_1$. This is the first demonstration that the R=20 constraint *structure* is shapeable by context choice rather than fixed. It does not by itself reduce the constraint count.

2.4 The R=19 Iteration Architecture Does Not Transfer Directly

The R=19 attack does not schedule its constraints acyclically: it too has a cycle (C0 requires W_9, which requires a_2, a_3, which come from C1/C2, which require a_1 from C0). It breaks the cycle by freezing W_9 at a provisional value, iterating C1$\leftrightarrow$C2 to a fixed point, and paying one 32-bit self-consistency check. Convergence of that iteration is what satisfies C1 and C2.

Promoting a_{11} from guessed context to an unknown absorbed by C3 gives R=20 the same one-word surplus (5 unknowns $a_0..a_3, a_{11}$ against 4 constraints, versus R=19's 4 against 3), and we verified that the true R=20 solution is a fixed point of all four absorber maps. The architecture therefore transplants in form. It does not transplant in efficacy. Measured with one harness, cold start, 8 iterations, no seeding, on the same hardware:

iteration	convergences / trials	rate per a_0
R=19 control (two-map, C1$\leftrightarrow$C2)	1,790 / 33,554,432	5.34×10^{-5}
R=20 (four-map, C0–C3)	0 / 1,996,488,704	$\leq 1.51 \times 10^{-9}$ (95%)

The R=19 control gives 5.34×10^{-5} per a_0. The main paper documents 1.3–1.9×10^{-5} per C0 survivor cold-start, i.e. 0.8–1.2×10^{-5} per a_0, and about 3.3× more with seeding [1, §5.4]; the control therefore sits within a factor of five of the cold-start figure and close to the seeded one. The harnesses differ in iteration cap and deduplication, and the R=20 conclusion below does not depend on which figure is taken. The cost accounting is as follows. Per context the iterated variant spends $2^{32} \cdot 8 = 2^{35}$ steps and yields $2^{32}p$ convergences, each passing the final 32-bit check with probability 2^{-32} under the uniform model, hence p preimages per context and $2^{35}/p$ steps per preimage; the classic harness spends 2^{32} per guessed a_{11} over 2^{32} guesses, i.e. 2^{64} per context-equivalent. The break-even convergence rate is therefore $p = 2^{-29} \approx 1.86 \times 10^{-9}$, and the measured bound lies below it. We note the margin is modest—the 95% bound is a factor 1.24 under break-even—so the conclusion is that the iterated variant does not beat the classic sweep, not that it loses by a wide factor; a longer run would tighten the bound proportionally. The structural estimate is far more pessimistic than the bound: R=19's 5.34×10^{-5} sits about 2^{15} above the 2^{-29} random-map baseline for a 32-bit state, and carrying that same enhancement to a 64-bit state (baseline $\approx 2^{-61}$) predicts $p \approx 2^{-46}$, some five orders of magnitude below break-even. The cause is dimensional: at R=19 the absorbed a_1 is computed *outside* the loop, leaving a 32-bit iterated state (a_3, with a_2 derived), whereas at R=20 C0 depends on a_{11} and the state becomes 64-bit (a_3, a_{11}). Raising the iteration budget does not help, since convergence probability and cost scale alike. Section 3.3 gives the structural reason the state cannot be held at 32 bits in *any* frame: at R=20 the full-avalanche dependency graph among the absorbed words acquires a cycle, and a cycle with no Maj/Ch-only edge has no cheap cut.

2.5 A Second Global Table: the a_0 Absorber

The attack sweeps a_0 and uses C0 to absorb a_1 through the global $\sigma_0(u) - u$ table. The two roles can be exchanged. Using

$$W_1 = a_1 + g(a_0), \qquad W_0 = a_0 + c_0, \qquad W_9(a_1) = W_9^{(0)} - a_1,$$

where $W_9^{(0)}$ is W_9 evaluated at $a_1 = 0$, the constraint $W_{16} = \sigma_1(W_{14}) + W_9 + \sigma_0(W_1) + W_0$ rearranges to give:

Proposition 1. *Let $\Gamma(a_0) := a_0 + g(a_0)$ and $R := W_{16} - \sigma_1(W_{14}) - W_9^{(0)} - c_0$. Then C0 is equivalent to*

$$\Gamma(a_0) \;=\; R + W_1 - \sigma_0(W_1).$$

The point is that g depends only on a_0 and the SHA-256 IV—no context word and no target enters it—so Γ is a fixed function and Γ^{-1} is a *global* table, built once and valid for every target and context, exactly like $\sigma_0(u) - u$. Measured over all 2^{32} inputs, Γ covers 63.214% of the range against $1 - e^{-1} = 63.212\%$ for a random map, so the table behaves like the one already in use. Proposition 1 is verified on the three preimages of [1, Table 2] by `code/verify_a0_absorber.py`, which also confirms it is not a tautology: perturbing W_{16} by $1, \ldots, 64$ breaks it in every case.

This does not reduce the attack's cost. C0 can absorb a_0 *or* a_1, but not both: exchanging the roles gives a mirror frame that sweeps W_1 and absorbs a_0, with the same number of absorbed words and the same dependency structure. What the second table changes is the inventory of §3.1, not the count; why the count cannot be improved is the edge-weight argument of §3.3.

2.6 A Linear Identity for σ_0, and its Destruction

Since σ_0 is $\mathbb{F}_2$-linear, put $L = \sigma_0 \oplus I$. The kernel of L is the fixed-point set $\{0, \texttt{0x27f42515}\}$, so $\operatorname{rank} L = 31$ and the image is a 31-dimensional subspace. Consequently there is a *unique* nonzero mask annihilating it.

Proposition 2. *Let $\lambda =$ 0xa8a42fe4. Then for every $W \in (\mathbb{Z}/2\mathbb{Z})^{32}$,*

$$\lambda \cdot \big(\sigma_0(W) \oplus W\big) = 0.$$

Verified exhaustively over all 2^{32} inputs (zero violations). This is the left companion of the known right-kernel constant `0x27f42515`.

The identity is however destroyed by modular subtraction, which is what the attack actually uses. Exhaustively over all 2^{32} inputs,

$$\Pr_W\big[\lambda \cdot (\sigma_0(W) - W) = 1\big] = 0.500017190,$$

a bias of $1.72 \times 10^{-5} = 2^{-15.8}$, against an exact 0 for the XOR form. A relation that is identically zero on the $\mathbb{F}_2$-linear part survives modular subtraction only as a $2^{-15.8}$ bias, which affords no usable filtering. Together with the table coverage—the σ_0 table covers 63.37% of its range, the Γ table 63.214% and the C3 h-table 63.21%, against $1 - e^{-1} = 63.212\%$ for a random map—the map $W \mapsto \sigma_0(W) - W$ is indistinguishable from a random map under every structural test we have applied. We conjecture that the interaction between $\mathbb{F}_2$-linear σ_0 and modular arithmetic is not merely an obstacle to proof technique but the mechanism by which R=20 resists.

The R=20 preimage attack remains an open problem.

3 The Absorber Assignment, and the Heavy-Cycle Barrier

This section gives the sharpest statement of the barrier we have, in two steps: an inventory of which absorptions admit a *global* table, and a measurement of the dependency structure among the absorbed words that explains, with one number per edge, why the R=19 iteration architecture cannot be carried to R=20 in either admissible frame. Neither step is an argument about cycles alone: the R=19 assignment is also cyclic and succeeds. What matters is the *weight* of the edges.

3.1 Linear Entries and the Global Absorbers

Write $T_2(x, y, z) = \Sigma_0(x) + \mathrm{Maj}(x, y, z)$. In the a-only representation of [1], $e_r = a_{r-4} + a_r - T_2(a_{r-1}, a_{r-2}, a_{r-3})$ and the recovered message word is

$$W_r \;=\; a_r - T_2(a_{r-1}, a_{r-2}, a_{r-3}) - e_{r-4} - \Sigma_1(e_{r-1}) - \mathrm{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) - K_r.$$

Lemma 2 (Linear entries)**.** *For every state word a_k: $W_k = a_k + f(a_{k-8}, \ldots, a_{k-1})$; $W_{k+8} = -a_k + g(a_{k+1}, \ldots, a_{k+8})$; a_k enters $W_{k+1}, \ldots, W_{k+7}$ nonlinearly; and it enters no other W_r.*

Proof. W_r depends on $a_{r-8}, \ldots, a_r$ and nothing else (the nine-word dependence of [1]). In W_k the word a_k occurs once, as the leading term. In W_{k+8} it occurs once, inside the h-term $e_{k+4} = a_k + a_{k+4} - T_2(a_{k+3}, a_{k+2}, a_{k+1})$, which is subtracted. For $1 \le m \le 7$ the dependence is nonlinear: a_k occurs inside a T_2, a Σ_1 or a Ch argument, and for $m = 4$ additionally as $-a_k$ through the h-term e_k, but in no case is the net entry linear. $\square$

The lemma is checked numerically by `edge_weights.py`: for $k \le 4$ at R=20 and four random increments each, W_k moves by exactly $+\delta$, W_{k+8} by exactly $-\delta$, the words between by neither, and $W_k + W_{k+8}$ is invariant under a_k.

Corollary 1 (Every constraint has a global absorber)**.** $\mathrm{C}_j = W_{16+j} - \sigma_1(W_{14+j}) - W_{9+j} - \sigma_0(W_{1+j}) - W_j$ *contains a_{j+1} exactly twice: as $+a_{j+1}$ inside the σ_0 argument W_{1+j} and as $-a_{j+1}$ inside W_{9+j}. With $u = W_{1+j}$ it reads $\sigma_0(u) - u =$ (terms free of a_{j+1}), so the single global table absorbs a_{j+1} from C_j for every $j = 0, \ldots, R - 17$.*

For $j = 0, 1, 2$ this is the opposite-sign pattern that drives the R=19 attack. For $j = 3$ it is C3$\to a_4$. The inventory of absorptions with the tables we know is:

Global (built once, valid for every target and context)	absorbs
$\sigma_0(u) - u$	$\mathrm{C}_j \to a_{j+1}$ for every j: C0$\to a_1$, C1$\to a_2$, C2$\to a_3$, C3$\to a_4$
Γ^{-1}, where $\Gamma(a_0) = a_0 + g(a_0)$ (§2.5)	C0$\to a_0$
$W_0 = a_0 + \text{const}$ (linear)	—
Per-context (2^{32} to build, reusable within one context)	
$Q_0^{-1}, Q_1^{-1}, Q_2^{-1}, H^{-1}$	C0–C3 $\to a_{11}$
No table (context-dependent inversion, 2^{32} every time)	
C3 absorbing a_1, a_2 or a_3; C0 absorbing a_4	—

3.2 Two Frames, Both Cyclic

Reaching 2^{32} per context requires sweeping one unknown and absorbing four. Given the tables we know, and up to the a_0/a_1 mirror of §2.5, there are exactly two ways to do it at R=20. *Frame A* (used in §2.4) keeps $a_4, \ldots, a_{10}$ as context, sweeps a_0, and absorbs a_1, a_2, a_3 globally and a_{11} per-context:

$$a_1 \leftarrow \text{C0 } (a_2, a_3, a_{11}), \quad a_2 \leftarrow \text{C1 } (a_1, a_3, a_{11}), \quad a_3 \leftarrow \text{C2 } (a_1, a_2, a_{11}), \quad a_{11} \leftarrow \text{C3 } (a_1, a_2, a_3).$$

Frame B keeps $a_5, \ldots, a_{11}$ as context, sweeps a_0, and absorbs $a_1, \ldots, a_4$ with the one global table:

$$a_1 \leftarrow \text{C0 } (a_2, a_3, a_4), \quad a_2 \leftarrow \text{C1 } (a_1, a_3, a_4), \quad a_3 \leftarrow \text{C2 } (a_1, a_2, a_4), \quad a_4 \leftarrow \text{C3 } (a_1, a_2, a_3).$$

Sweeping a_1 instead of a_0 gives the mirror frame of §2.5 (sweep W_1, absorb a_0 through Γ^{-1}) and changes nothing below; sweeping a_2 or a_3 leaves a_0 and a_1 both to be absorbed by C0, which cannot absorb two words. Both assignments are cyclic. So is the R=19 assignment ($a_1 \leftarrow \text{C0}(a_2, a_3)$, $a_2 \leftarrow \text{C1}(a_1, a_3)$, $a_3 \leftarrow \text{C2}(a_1, a_2)$), and it succeeds; cyclicity is not the barrier.

3.3 Edge Weights: the Heavy Cycle Appears at $R = 20$

What distinguishes the three assignments is how strongly each absorbed word feeds back into the constraints that absorb the others. We measure it directly: for random targets, contexts and unknowns, flip one random bit of a_k and record the Hamming weight of the change in C_j, averaged over 200 trials per cell (`edge_weights.py`, seed 20260902; raw output in `data_edge_weights.txt`). Full avalanche is about 16; a *mild* edge, one that passes only through Maj and Ch of a_k plus a constant, moves one or two bits. We call a feedback edge *heavy* when it moves more than 4 bits; self-edges sit near 8 because the absorbed word moves its own $\sigma_0(u) - u$ term.

	R=19			R=20, frame B				R=21				
flip	C0	C1	C2	C0	C1	C2	C3	C0	C1	C2	C3	C4
a_0 (swept)	15.4	15.9	15.9	15.6	15.8	15.8	15.0	15.1	16.1	15.7	15.2	15.5
a_1	<u>8.0</u>	15.3	16.2	<u>7.6</u>	15.2	16.1	15.7	<u>8.0</u>	15.1	16.2	15.9	15.3
a_2	1.5	<u>7.9</u>	15.1	1.3	<u>7.9</u>	14.9	16.3	1.5	<u>7.6</u>	14.9	16.1	16.0
a_3	1.9	1.4	<u>8.3</u>	1.9	1.6	<u>8.3</u>	15.2	1.7	1.4	<u>7.8</u>	15.4	16.1
a_4				**12.2**	1.7	1.4	<u>7.8</u>	**12.4**	1.8	1.5	<u>7.8</u>	15.2
a_5								**8.3**	**12.4**	1.6	1.3	<u>8.0</u>

Underlined cells are self-edges (the absorbed word moving its own $\sigma_0(u) - u$ term); entries above the diagonal are forward edges; entries below it are feedback edges, with the heavy ones in bold. Frame A at R=20 (rows a_1, a_2, a_3, a_{11}, same script) has feedback weights $a_{11} \to$C0 15.7, $a_{11} \to$C1 13.3, $a_{11} \to$C2 15.0.

Reading. At R=19 every feedback edge is mild. The full-avalanche ("heavy") dependency graph is the chain $a_1 \to a_2 \to a_3$, which is acyclic, and the attack exploits exactly this: it computes a_1 outside the loop, iterates $a_3 \to a_2 \to a_3'$, and thereby cuts its cycle at the mild edge $a_3 \to$C1. That is why the iterated map is nothing like a random map (its convergence rate of 5.34×10^{-5} sits 2^{15} above the random-map baseline, §2.4). At R=20 exactly one feedback edge is heavy, $a_4 \to$C0 at 12.2 bits, and the heavy graph acquires the cycle $a_1 \to a_2 \to a_3 \to a_4 \to a_1$. A heavy cycle has no mild cut. Cutting it at any single 32-bit word composes full-avalanche maps with table lookups, which behaves, as far as we can measure, like a random map on 2^{32} points; its fixed points then cost 2^{32} each to find, which is the classic sweep. Frame A is worse, with three heavy feedback edges through a_{11}. This is the content of the "dimensional" explanation in §2.4: the state cannot be held at 32 bits because no mild cut exists.

Why it is the round function and not the frame. The weight of an entry is set by its offset. a_k enters W_{k+m} for $m = 6, 7$ only through Maj and Ch of a_k plus constants (mild); for $m = 5$ through the h-term e_{k+1}, which contains $\Sigma_0(a_k)$ subtracted directly, and through $\Sigma_1(e_{k+4}) = \Sigma_1(a_k + c)$ (heavy; measured 12.2–12.4); for $m = 4$ through $\mathrm{Ch}(e_{k+3}, e_{k+2}, e_{k+1})$, which selects about half the bits of $\Sigma_0(a_k)$ (measured 8.3); and for $m \leq 3$ through Σ_0 and Σ_1 of a_k itself. The last word of the window, a_n with $n = R - 16$, feeds C0 through W_9 at offset $9 - n$, which is mild if and only if $n \leq 3$, i.e. $R \leq 19$. No context choice removes $\Sigma_0(a_4)$ from W_9: it enters additively with coefficient $+1$ through the subtracted h-term e_5, and a_4 reaches C0 through no other word, since W_0, W_1, W_{14} and W_{16} all lie outside offsets 0–8 from a_4. At R=21 the window has three heavy feedback edges (table), and the count grows from there.

What would cut it. A trick that cancels the barrier must cut one heavy edge, and the cheapest candidate is the new one: in C0, invert $\Sigma_0(a_4) - \Sigma_1(a_4 + c_8)$ plus its Maj/Ch terms *jointly* with the a_1 absorber $\sigma_0(W_1) - W_1$ at well below 2^{32} per instance. All of these are rotation–XOR maps, so over GF(2) they would combine linearly; the modular carries between them are what destroy the combination, and §2.6 measures that destruction for σ_0 alone (an exact GF(2) identity degraded to a $2^{-15.8}$ bias). Other routes one might try—a normal form for $\sigma_0(W_4) + W_3$ in one message word, freeing the sweep variable through the a_0 absorber, a joint two-variable global table—are all instances of cutting a heavy edge, and none has been found. The two attempts recorded below are of this kind and both charge full price.

3.4 Attempts to Cut the Cycle

The one structural handle, and why it does not pay. C3 has the form $\sigma_0(W_4) + W_3 = H(a_{11})$, coupling *two* message words—this is precisely why it cannot absorb a_1, a_2, a_3 through the single-word table. Imposing any global relation $W_3 = \phi(W_4)$ collapses the left side to a function of one word, making C3 a global absorber producing (W_3, W_4) from a_{11} alone and reversing the cycle direction. Measured exhaustively in the reduced-width model at $w = 4$ over 6 contexts (98 true solutions; expectation 6.12 if ϕ costs a full w bits):

Relation	surviving	per context	vs. expected
$W_3 = \sigma_0(W_4)$	3	0.50	0.49×
$W_3 = -W_4$	4	0.67	0.65×
$W_3 = W_4$	0	0.00	0.00×
$W_3 = \overline{W_4}$	7	1.17	1.14×

Every variant sits at or near the "costs a full w bits" expectation. The counts are small (3–7 events, so roughly ±2) and all are consistent with 1.0×; none is consistent with the large excess needed to make the handle pay. Imposing a w-bit condition removes a w-bit worth of solutions, and the global absorber it buys does not compensate.

Elimination orders. An elimination order cannot change an edge weight, so the heavy-cycle argument predicts that no order recovers the solution set. The exhaustive search over all 2,880 elimination orders in the reduced-width model (w=4, two contexts, 41 true solutions) agrees: 2,560 of 2,880 orders find at least one solution, the median finds 2 and the best finds 11, and the hand-chosen R=19-style order (sweep a_0; $a_1 \leftarrow$C0, $a_2 \leftarrow$C1, $a_3 \leftarrow$C2, $a_{11} \leftarrow$C3) ranks 175th with 5. No order recovers more than 27% of the solution set. The search treats every absorption as available, ignoring the table inventory, so it is if anything generous to the alternatives.

4 Status

$R = 20$ is not solved. The barrier is 2^{64} per context-equivalent for this attack family, 2^{32} times the R=19 per-context cost, or about $2^{71.4}$ expected total work at the projected R=19 rate, and its sharpest form is the heavy-cycle argument of §3.3: the full-avalanche dependency graph among the absorbed words is acyclic for $R \leq 19$ and cyclic from R=20 in every admissible frame, because $\Sigma_0(a_k)$ is subtracted directly into W_{k+5}. Context screening ([1, §5.7]) raises W_9 lo-pass yield by a measured 3.41× (95% CI $[2.13, 5.82]$) with conversion to preimages unmeasured; taken at face value that lowers the expected total work by the same factor, to about $2^{69.6}$, an improvement and not a solution.

We emphasise the limits of the negative results. Each rules out a parameterisation, not a technique: the ϕ-family charges a full word for the global absorber it buys; a_0 is globally absorbable but competes with a_1 for C0; no elimination order recovers more than a quarter of the solution set. The heavy-cycle argument is a measurement of this attack family, not a lower bound. What would change the picture is a way to invert $\Sigma_0(a_4) - \Sigma_1(a_4 + c_8)$ jointly with $\sigma_0(W_1) - W_1$ inside C0 at well below 2^{32}, and nothing measured here excludes one; it does say that any such trick is a statement about carries between rotation–XOR maps, which is where §2.6 found only a $2^{-15.8}$ bias.

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